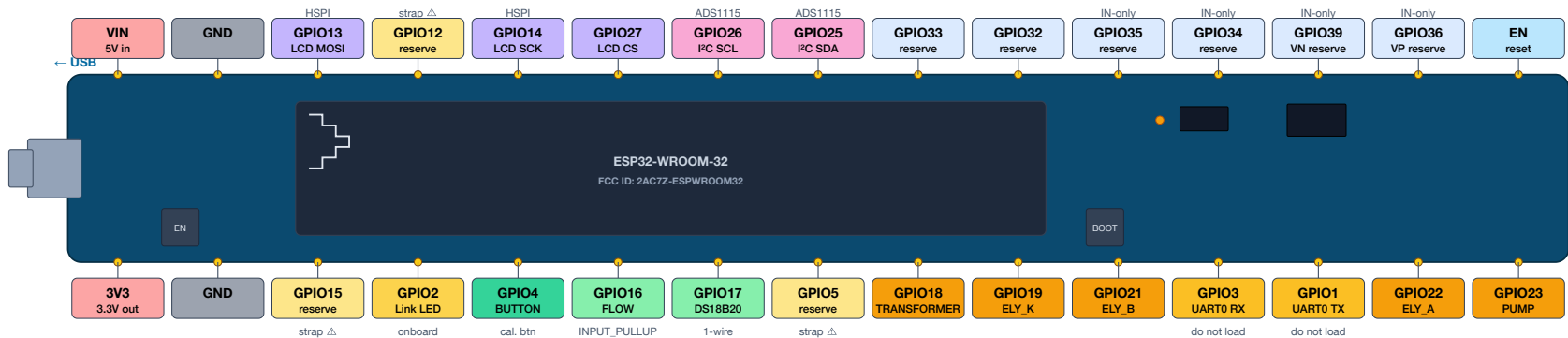


Poolcontroller09 · ESP32-WROOM-32 (30-pin DevKit) · USB ←

Project-specific pinout · orientation: USB on left

Board drawn with USB connector on the left. All actuators, sensors, flow switch and calibration button live on the BOTTOM row (closer to the user); display (HSPI) + I²C (ADS1115) live on the TOP row.



Project wiring (bottom row)

GPIO	Function	Direction	Notes
GPIO4	Calibration button	INPUT_PULLUP	Button to GND
GPIO16	Flow switch	INPUT_PULLUP	Active LOW
GPIO17	DS18B20 data	IN/OUT	4.7 kΩ → 3V3
GPIO18	Transformer relay	OUT	Active HIGH (auto)
GPIO19	Electrolysis K relay	OUT	Active HIGH
GPIO21	Electrolysis B relay	OUT	Active HIGH
GPIO22	Electrolysis A relay	OUT	Active HIGH
GPIO23	Pump relay	OUT	Active HIGH
GPIO1 / GPIO3	UART0 (serial log)	—	do not load
GPIO0 / GPIO2 / GPIO5 / GPIO15	Strap reserves	—	leave open

Top row – display + I²C

GPIO	Function	Notes
GPIO13	ST7920 MOSI	HSPI MOSI
GPIO14	ST7920 SCK	HSPI CLK
GPIO27	ST7920 CS	idle HIGH
GPIO25	ADS1115 SDA	I²C, 4.7 kΩ → 3V3
GPIO26	ADS1115 SCL	I²C, 4.7 kΩ → 3V3
GPIO12	reserve (strap)	MUST be LOW at boot
GPIO32 / 33	reserve	in/out available
GPIO34 / 35 / 36 / 39	reserve	input-only (no pull)

Voltage compatibility (excerpt)

Device	Note
DS18B20 / ADS1115 / button / flow	3.3 V native, OK directly
ST7920 LCD	5 V module — 3.3 V signals marginal ($V_{IH} \geq 3.5$ V). Use 3.3 V-logic module, or level-shift SCK/MOSI/CS via 74HCT125 / TXS0108E.
5 V relay boards	Many opto-relay boards need ≥ 4 V IN. Prefer 3.3 V-logic board, or add 74HCT125 5 V buffer.

☐ Power ☐ GND ☐ Reset / EN ☐ GPIO reserve ☐ Strap pin (do not load) ☐ UART0 (serial log) ☐ Relay output ☐ Sensor in ☐ Button ☐ SPI (LCD) ☐ I²C (ADS1115)
☐ Onboard LED

Generated from `scripts/pinout.html`. Source pin map in `src/main.cpp`. Reference board layout based on DOIT ESP32-DEVKIT V1 (30-pin). Your 30-pin board should match; if not, flip the row/direction in the SVG script.